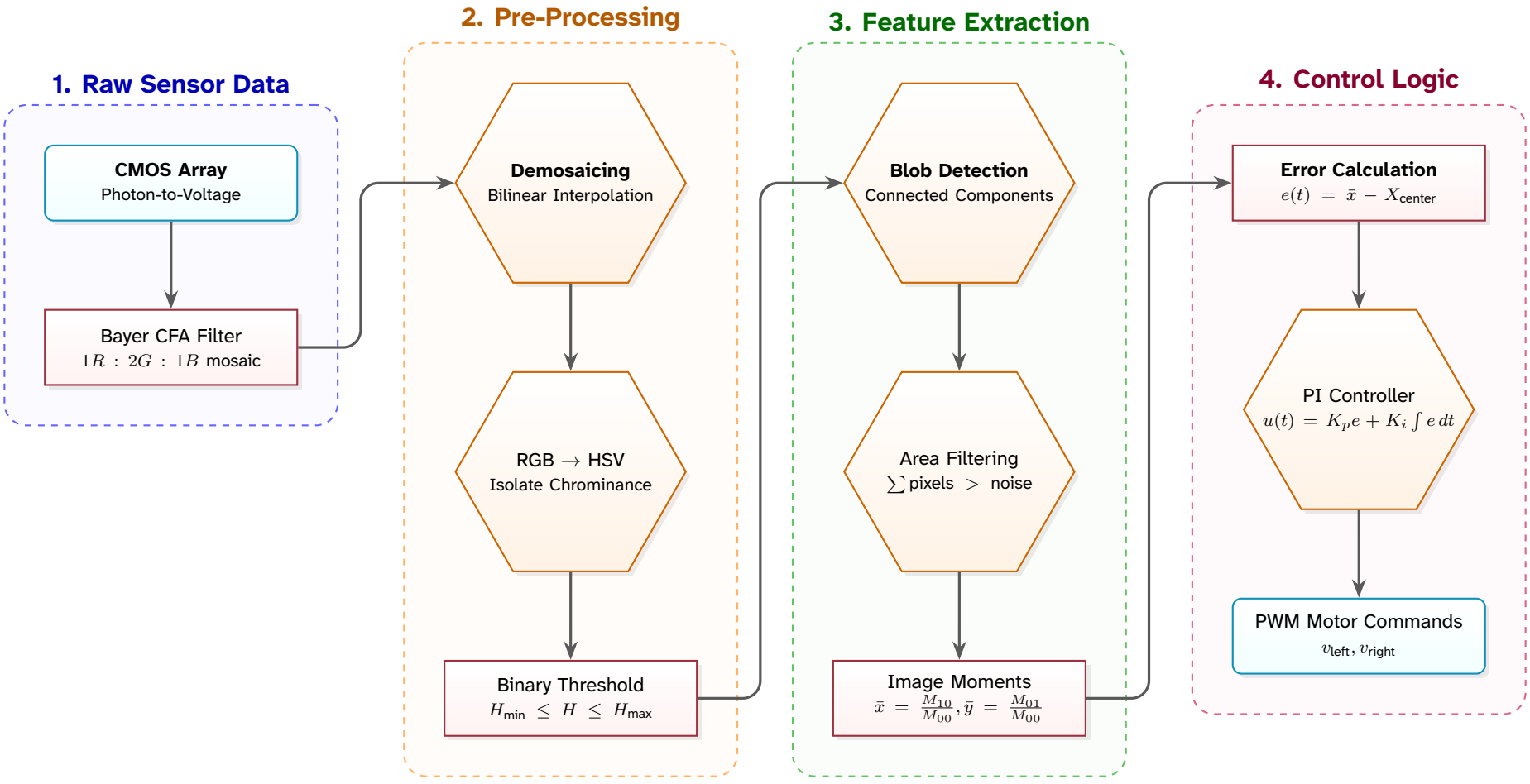

1. Raw Sensor Data
CMOS Array
Photon-to-Voltage
Bayer CFA Filter
$1R : 2G : 1B$ mosaic
2. Pre-Processing
Demosaicing
Bilinear Interpolation
RGB → HSV
Isolate Chrominance
Binary Threshold
$H_{\text{min}} \leq H \leq H_{\text{max}}$
3. Feature Extraction
Blob Detection
Connected Components
Area Filtering
$\sum$ pixels $>$ noise
Image Moments
$\bar{x} = \frac{M_{10}}{M_{00}}, \bar{y} = \frac{M_{01}}{M_{00}}$
4. Control Logic
Error Calculation
$e(t) = \bar{x} - X_{\text{center}}$
PI Controller
$u(t) = K_p e + K_i \int e\, dt$
PWM Motor Commands
$v_{\text{left}}, v_{\text{right}}$